

Recommendations — mcrc-kras-a59t-pik3ca-peritoneal-pre-hi pec-q8k4

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** colon
- **Histology:** colon adenocarcinoma (presumed; histologic subtype not specified at intake)
- **Stage:** IV (metastatic; liver metastases status post-resection, peritoneal involvement present)
- **ECOG:** 1
- **Age band:** 30-39
- **Sex:** unknown
- **Biomarkers:** Germline hereditary cancer panel negative; KRAS A59T mutated (atypical, switch-II region; non-G12/G13/Q61) (ngs_pending); APC E1295 mutated (likely truncating; presumed mutation cluster region) (ngs_pending); TP53 R273 mutated (hotspot; specific substitution not specified) (ngs_pending); PIK3CA M1043I mutated (kinase-domain activating) (ngs_pending); SMAD4 R361H mutated (MH2-domain hotspot; loss-of-function) (ngs_pending); 20q11 amplification (BCL2L1/TOP1 co-amplified) amplified (ngs_pending); MSI status MSS; TMB low; PD-L1 negative; HER2 (ERBB2) amplification not reported (ngs_pending); BRAF V600E not explicitly reported (ngs_pending); NRAS / HRAS not explicitly reported (ngs_pending); Consensus molecular subtype (CMS) / CDX2 not reported (ngs_pending)
- **Efficacy/toxicity weight:** 0.7

4 ranked therapeutic options across 3 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Pik3Ca M1043I interventions

1. Low-dose aspirin 160 mg PO daily as a PIK3CA-directed adjunct, sequenced into the post-HIPEC window and held peri-operatively

Likelihood of effect: Moderate as a recurrence-risk adjunct: ALASCCA HR 0.49 in the exon-9/20 subset M1043I maps into; direction concordant across three datasets, but validated in resected not metastatic disease.

Toxicity burden: Low (G3+ GI bleeding ~3%; serious AEs ~5%)

Counter-productive MoA:Low (Additive bleeding risk with concurrent bevacizumab and the CRS-HIPEC window; a sequencing/timing hold, not a mechanism that blunts the therapeutic goal.)

Overall:The strongest RoB-low biomarker match and a do-no-harm PIK3CA-directed adjunct; effect validated in resected, not metastatic, disease, so it layers on the backbone rather than replacing it.

Key references: PMID 40961426 · PMID 38889377 · PMID 23094721

2. Mutant-selective PI3K-alpha inhibition - CGT6297 on the explicit CRC + PIK3CA cohort (NCT07383506, Cohort 3), with tersolisib (NCT05768139) / ETX-636 (NCT06993844) / inavolisib+bev on INTRINSIC (NCT04929223) as class alternatives

Likelihood of effect: Low-to-moderate for CRC: cross-tumor kinase-domain-PIK3CA signal (INAVO120 HR 0.43 in breast; RLY-2608 ORR 66.7% breast) with only 2 alpelisib CRC PRs and a non-canonical-allele hedge.

Toxicity burden: Moderate (hyperglycemia G3+ ~20% alpelisib / ~6% inavolisib, stomatitis, rash, diarrhea; near-absent on wild-type-sparing agents)

Counter-productive MoA: **Moderate** (KRAS A59T co-activation can bypass PI3K-alpha blockade via parallel MAPK signaling, blunting single-agent response and arguing for combination.)

Overall: The one trial naming a CRC-and-PIK3CA cohort, targeting the patient's kinase-domain allele directly; efficacy rests entirely on cross-tumor abstract-tier data with a non-canonical-allele predictive hedge.

Key references: PMID 39476340 · PMID 29401002 · NCT07383506

Evidence in detail

Low-dose aspirin 160 mg PO daily as a PIK3CA-directed adjunct, sequenced into the post-HIPEC window and held peri-operatively

Frödin et al. *N Engl J Med* 2025

- **Disease context:** Resected stage I-III rectal or stage II-III colon cancer with somatic PI3K-pathway alteration (PIK3CA / PIK3R1 / PTEN) (adj) — n=626 randomized; 1,103 of 2,980 screened (37%) carried a PI3K-pathway alteration. Group A = PIK3CA exon 9/20 mut; Group B = other PI3K-pathway alterations.
- n: 626
- **Toxicities:** GI bleeding (grade ≥3): 3%; **Serious AEs overall** (grade any): 5%
- **Efficacy:** HR_RFS 0.49 (95% CI 0.24-0.98); HR_RFS 0.42 (95% CI 0.21-0.83); RFS 92.3% at 3 yr; RFS 85.9% at 3 yr

Yang et al. *J Clin Oncol* 2024

- **Disease context:** Resected stage III colon cancer, PIK3CA-activating subset (adj) — Pre-specified biomarker subgroup of CALGB/SWOG 80702 (n=2,524 randomized total; ~16% PIK3CA-activating).
- n: 401
- **Toxicities:** Celecoxib G3+ AEs class-typical (cardiovascular, GI). Cohort safety dominated by FOLFOX backbone toxicity.
- **Efficacy:** HR_DFS 0.56 (95% CI 0.35-0.88); HR_OS 0.41 (95% CI 0.21-0.79); HR_DFS 1.07 (95% CI 0.89-1.28)

Liao et al. *N Engl J Med* 2012

- **Disease context:** Resected colorectal cancer stratified by tumor PIK3CA mutation (adj) — 964 stage I-IV CRC patients from NHS + HPFS; tumor PIK3CA mutation status assessed retrospectively; 17% PIK3CA-mut.
- n: 964
- **Toxicities:** Aspirin GI-bleed / ulcer risk class-typical; not formally captured in this cohort analysis.
- **Efficacy:** HR_OS 0.18 (95% CI 0.06-0.61); HR_OS 0.54 (95% CI 0.31-0.94); HR_OS 0.96 (95% CI 0.69-1.32)

Mutant-selective PI3K-alpha inhibition - CGT6297 on the explicit CRC + PIK3CA cohort (NCT07383506, Cohort 3), with tersolisib (NCT05768139) / ETX-636 (NCT06993844) / inavolisib+bev on INTRINSIC (NCT04929223) as class alternatives

****Jhaveri et al. *N Engl J Med* 2024****

- **Disease context:**HR+/HER2- PIK3CA-mutated metastatic breast cancer (1L after endocrine therapy) (1L) — n=325 with PIK3CA alterations confirmed by ctDNA or tissue.
- **n:** 325
- **Toxicities:**Hyperglycemia (grade ≥3): 5.6%; Hyperglycemia (grade any): 58.6%; Stomatitis (grade ≥3): 5.6%; Diarrhea (grade ≥3): 3.7%; Rash (grade any): 25%; Treatment discontinuation due to AE (grade any): 6.5%
- **Efficacy:**HR_PFS 0.43 (95% CI 0.32-0.59); PFS 15.0 months; PFS 7.3 months; ORR 58%; ORR 25%

****Juric et al. *J Clin Oncol* 2018****

- **Disease context:**Advanced solid tumors with PIK3CA alterations (mixed: breast, CRC, gynecologic, endometrial) (2L+) — n=134 PIK3CA-altered solid tumors; CRC subset included. PIK3CA mutations spanned the canonical helical / kinase hotspots (E542K, E545K, H1047R) and rarer alleles.
- **n:** 134
- **Toxicities:**Hyperglycemia (grade ≥3): 20%; Rash/maculopapular (grade ≥3): 10%; Diarrhea (grade ≥3): 7%; Nausea (grade any): 56%
- **Efficacy:**ORR 6%; RR 2 PRs

Mss Peritoneal interventions

1. Botensilimab + balstilimab (Fc-enhanced anti-CTLA-4 + anti-PD-1), 2L+ after FOLFIRI-bev progression, on an MSS-CRC / any-KRAS protocol (USC NCT06336902, Agenus NCT05608044, CCTG NCT07152821, or expanded access NCT06751524)

Likelihood of effect: Moderate in this NLM/peritoneal subset: ORR ~22% and mOS 20.9 mo (Bullock 2024), the strongest MSS-CRC ICI signal, but subgroup-level and not yet confirmed vs BSC.

Toxicity burden: Moderate (immune-mediated colitis 12-13%, hepatitis ~8%, pyrexia/cytokine-release; overall G3+ irAE ~35-38%)

Counter-productive MoA:Moderate (Fc-enhanced CTLA-4 irAE colitis/hepatitis overlaps the peri-operative CRS window; immune-mediated toxicity could delay or complicate cytoreductive surgery.)

Overall:The only MSS-CRC immunotherapy with a compartment-matched signal at the patient's peritoneal disease; a 2L+ proposition gated by ~35% immune toxicity and single-arm, subgroup-level evidence.

Key references: PMID 38871975 · NCT07152821 · NCT06336902

Evidence in detail

Botensilimab + balstilimab (Fc-enhanced anti-CTLA-4 + anti-PD-1), 2L+ after FOLFIRI-bev progression, on an MSS-CRC / any-KRAS protocol (USC NCT06336902, Agenus NCT05608044, CCTG NCT07152821, or expanded access NCT06751524)

****Bullock et al. *Nat Med* 2024****

- **Disease context:** Relapsed/refractory MSS metastatic colorectal cancer (2L+) — n=101 r/r MSS mCRC, mixed liver / non-liver metastatic distribution. Most patients had ≥3 prior lines.
- **n:** 101
- **Toxicities:** Diarrhea/colitis (grade ≥3): 12%; Fatigue (grade any): 45%; Pyrexia (grade any): 38%; Hepatitis (transaminitis) (grade ≥3): 8%; Treatment-related AE ≥3 (grade ≥3): 35%; Treatment-related death (grade 5): 0% (n=101)
- **Efficacy:** ORR 17%; ORR 22%; ORR 5%; OS 20.9 months; DoR NR months

Kras A59T interventions

1. Daraxonrasib (RMC-6236) and other codon-agnostic pan-RAS / RAS(ON) inhibitors on trials whose eligibility does not name a codon (Pfizer PF-07934040 NCT06447662 'any KRAS variant', Roche RO7673396 NCT06884618, Jacobio JAB-23E73 NCT06973564, Adlai AN9025 NCT07252479, RevMed RAS(ON)+ivonescimab NCT07397338), contingent on a sponsor A59T-eligibility call

Likelihood of effect: Low / unquantified for A59T: zero allele-matched data; extrapolated from PDAC canonical-codon ORR 35% (pmid:42090791) and structural binding (pmid:40056080, A59T untested); CRC monotherapy signal ~9%.

Toxicity burden: Moderate (daraxonrasib G3+ TRAE ~30%: rash, stomatitis/mucositis, diarrhea; combination arms uncharacterized in CRC)

Counter-productive MoA: Moderate (Switch-II A59T may signal weakly through MAPK, so pan-RAS engagement optimized for G12/G13 could under-suppress the pathway; VEGF-arm wound-healing impairment peri-HIPEC.)

Overall: The only option reaching the patient's actual KRAS A59T driver, but on codon-agnostic eligibility and structural read-across with zero allele-matched efficacy; a sponsor-gated trial referral, not proven A59T therapy.

Key references: PMID 42090791 · PMID 40056080 · NCT06447662

Evidence in detail

Daraxonrasib (RMC-6236) and other codon-agnostic pan-RAS / RAS(ON) inhibitors on trials whose eligibility does not name a codon (Pfizer PF-07934040 NCT06447662 'any KRAS variant', Roche RO7673396 NCT06884618, Jacobio JAB-23E73 NCT06973564, Adlai AN9025 NCT07252479, RevMed RAS(ON)+ivonescimab NCT07397338), contingent on a sponsor A59T-eligibility call

****Wolpin et al. *N Engl J Med* 2026****

- **Disease context:** Previously treated advanced RAS-mutated pancreatic ductal adenocarcinoma (2L+) — RAS-mutant PDAC; efficacy anchored to the 300 mg second-line RAS G12 subset (n=26) and the G12/G13/Q61 subset (n=38). Codons 12/13/61 only — no A59 alleles enrolled.
- **n:** 168
- **Toxicities:** Treatment-related AE (grade ≥3): 30% (n=168); Rash (grade any); Stomatitis/mucositis (grade any); Diarrhea (grade any)

- **Efficacy:** ORR 35% (95% CI 17-56); ORR 29% (95% CI 15-46); PFS 8.5 months; OS 13.1 months; DoR 8.2 months

****Knox et al. *J Med Chem* 2025****

- **Disease context:** Preclinical characterization of a pan-RAS(ON) tri-complex inhibitor across RAS variants (any) — Cell-line and structural work spanning wild-type KRAS and G12V/G12C/G12D/G12R/G13D/Q61X. The compound occupies a CypA-SWI/SWII composite pocket and leaves the G12/G13/Q61 groove unoccupied, the structural basis for binding mutant and wild-type RAS with similar affinity.
- **n:** —
- **Toxicities:** —
- **Efficacy:biomarker** pERK reduction across 8 RAS-variant lines